Ivan Kirischian

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Website: bigbirdy7.github.io/myprojects

Experience

Math & Programming Tutor | Independent | May 2026 – Present

- Plan and deliver personalized lessons in math and programming for a private student
- Regularly tutor classmates and siblings informally after school

VEX V5 Junior Team Lead | Thornhill Secondary School Robotics Club | January 2026 – Present

- Dedicated more time than any other junior team member, including taking the robot home to work on independently
- Built a competition-ready robot in under a week for STEM Day, then taught visiting students how it works and how to drive it
- Represented the computer engineering program at the school Open House, explaining resources and demonstrating projects

Skills

- Robotics/Electronics: VEX V5/IQ, Arduino, Mbot, Microbit, LEGO Mindstorms
- Programming: Java, Python, C++
- Experienced with CAD tools like Tinkercad, SketchUp and Fusion360 (for VEX V5)
- Strong communication skills and patience when teaching others. My greatest soft skill is efficiently explaining complex concepts, from the ground up.

Awards

VEX Robotics Regional Competition | York Region | March 2026

- Competed with Thornhill Secondary School Robotics Club, placing second and advancing to provincials

McMaster Sumo Bot Competition, First and Third place | January 2025

- Contributed line detection programming to the first place robot
- Independently programmed a second school team's robot entirely solo, which placed third

Academic Awards

- Top of class in Grade 12 Computer Science, completed in Grade 11
- Top of class in Grade 11 Computer Science, completed in Grade 10

Personal Projects (bigbirdy7.github.io/myprojects)

- **3D Renderer and Flight Sim, From Scratch**
  Programmed a 3D renderer in Java, completely from scratch, without copying tutorials. The only libraries it uses are a library for coloring pixels on a window. I learned about linear algebra, rendering and rasterising, then used my knowledge to create the project. Later I added multiplayer functionality.
- **Robot Car with Laser Turret**
  Built a mobile robot with a laser turret controllable by a smartphone app, took apart servos and replaced inner components to achieve our project goal in a limited timeframe.
- **Real Life Portal Turret with AI vision (In-progress)**
  A 3D printed turret that tracks a person as they walk through the room and “shoots” at them with LED lights. I greatly improved my soldering skills when making the project..
- **Math Quiz Device for Kids**
  Created an educational game using an Arduino, keypad, and LCD to help my younger sister practice math, complete with a point system and a custom 3D-printed case.
- **Personal AI Assistant**
  Desktop AI assistant using Groq API, microphone input, and a text-to-speech model. Inspired by the character GLaDOS from Portal 2. I am planning to integrate it into the Real Life Portal Turret.

Education

Thornhill Secondary School – Grade 11 Student
Graduation: 2027

- Relevant Courses: Computer Science 98% (grade 12) Computer Science 97% (grade 11), Computer Engineering 97% (grade 11) Computer Engineering 100% (grade 10)
- Executive in Robotics Club (Will be president next year) and Coding Club

I am a high school student with a passion for robotics, coding, and technology, actively seeking opportunities to teach and mentor younger students. I am always eager to share my experience in designing, building, and programming robots and electronics projects. As a strong communicator and adaptable team player, my greatest soft skill is efficiently explaining complex concepts from the ground up. To apply this, I also tutor, teaching math and programming to kids, while also regularly tutoring my peers. I also enjoy volunteering; recently, I ran the junior VEX V5 section for my school's STEM day, teaching visiting students how to drive the robots.